Aqeel Ahmed, PhD

Physicist | Quantitative Researcher | Data Scientist

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Summary

PhD Physicist with 10+ years of experience in high-dimensional quantitative modeling, statistical analysis, and large-scale computational simulation. Expert in leading data-driven research initiatives and engineering complex analytical solutions using Python, C++, and Mathematica with a proven track record of 30+ publications. Applying this extensive quantitative expertise to deliver high-impact Data Science and Machine Learning/AI solutions.

Skills

Data Science & ML	NumPy, SciPy, Pandas, Scikit-learn, PyTorch, PySpark, SQL, Google Cloud
Programming & Tools	Python, C/C++, Mathematica, Git, Linux/Bash, LaTeX, Jupyter, Docker
Mathematics	Stochastic Modeling, Monte Carlo Simulations, Symbolic Analysis, Statistics
Soft Skills	Project Management, Communication, Critical Thinking, Team Collaboration
Languages	English (Fluent), German (Intermediate B1) [B2 in progress]

Experience (Postdoctoral Researcher)

- **Max-Planck-Institut für Kernphysik**, *Heidelberg, Germany* — Oct 2020 – Dec 2025
  - Architected high-dimensional predictive models, translating abstract mathematical theories into scalable, production-ready Python and C++ codebase.
  - Designed and optimized automated data processing pipelines for large-scale datasets, significantly reducing computational overhead and simulation runtimes.
- **Vrije Universiteit Brussel**, *Brussels, Belgium* — Oct 2018 – Sep 2020
  - Developed end-to-end analytical pipelines using Python (NumPy, Pandas) to automate the ingestion, cleaning, and statistical validation of complex experimental data.
  - Created robust data visualizations to translate highly technical quantitative findings into actionable insights.
- **Johannes Gutenberg University**, *Mainz, Germany* — Dec 2017 – Sep 2018
  - Engineered custom numerical algorithms in Python and Mathematica to solve high-dimensional optimization problems, accelerating analytical workflows.
- **University of Warsaw**, *Warsaw, Poland* — Apr 2016 – Nov 2017
  - Managed end-to-end quantitative research projects, performing rigorous statistical modeling and advanced mathematical simulations to validate complex hypotheses.

Education

◇ PhD Physics — Dec 2011 – Jan 2016
University of Warsaw, Warsaw, Poland — *International PhD Fellowship*
Visiting Scholar at UC Riverside (1 year) and UC Davis (6 months)

◇ M.Sc. & M.Phil Physics — Feb 2006 – Apr 2010
Quaid-i-Azam University, Islamabad, Pakistan — *Chancellor Gold Medal*

Leadership & Management

- **Mentorship & Team Lead:** Co-supervised and mentored 10+ PhD and Master's students, guiding technical implementation and ensuring project delivery against strict deadlines.
- **Technical Communication & Event Management:** Presented complex quantitative findings at 50+ international conferences; organized 10+ technical workshops, managing logistics and content for 200+ attendees.
- **Knowledge Transfer:** Taught advanced mathematics and physics courses at the Heidelberg University, VUB Brussels, University of Warsaw, and QAU Islamabad, translating highly abstract concepts for diverse audiences.
- **Quality Assurance (Data & Models):** Appointed technical referee for top-tier physics journals (JHEP, PRD, EPJC); rigorously verified mathematical validity, algorithmic logic, and data integrity.
- **Cross-Functional Alignment:** Acted as Postdoc Representative and Project Convener for the *be-h consortium Belgium*, aligning technical roadmaps and research goals across 5+ international institutions.